

Robustness of the Mean and Median under Normal Contamination

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Abstract

Under the pre-registered synthetic setting, the sample mean had RMSE 0.492, while the sample median had RMSE 0.178; the median reduced RMSE by 63.8%.

Design

We generated 200 samples of size 50 from a centered normal mixture with 10% contamination. Robust estimation is motivated by classical work (Huber 1964).

Results

Figure 1 reports the fixed-seed result (mean at left; median at right). These values describe this simulation only and do not establish universal superiority.

Limitations

This demo uses one distribution, one sample size, and one fixed simulation seed. The local literature record is an abstract-only test fixture, not a systematic review.

AI Use Disclosure

AI agents assisted workflow orchestration and drafting; all claims are bound to the audited local ledger.

Huber, Peter J. 1964. “Robust Estimation of a Location Parameter.” *The Annals of Mathematical Statistics* 35 (1): 73–101. <https://doi.org/10.1214/aoms/1177703732>.



Figure 1: Estimator RMSE